10.3.2 Text-to-speech integration

Integrating external services like Google Text-to-Speech requires an understanding of the technology and its integration into the application. Minimizing the latency of the text-to-speech functionality to provide a seamless user experience is also a key priority. As part of the solution, we implement a protected API route (`/api/tts`) that streams the audio content back to the client. This approach not only simplifies the feature delivery to the UI but also offloads the processing to the server.

We also use a simple feature flag support using the env variable to enable this feature, but it would be a more reasonable solution to have an external API or service that better controls those features. While using environment variables for feature flags provides a quick and straightforward way to toggle the text-to-speech functionality, this approach lacks the flexibility and control needed for more complex scenarios. In a production environment, a dedicated feature management service offers significant advantages, such as the ability to enable or disable features for specific user-facing segments.

10.3.3 Generating feedback

We use a simple prompt to guide the agent in providing feedback based on the interview thread:

```
const prompt = `
Please provide comprehensive feedback for this interview.
    The session contains the following messages:
    ${messages.map((message) => `${message.role}: ${message.content}`).join('\n')}

    Provide a structured feedback summary covering these points:
    - Overall performance: Assess the candidate's overall performance.
    - Strengths: Highlight the candidate's
strengths demonstrated during the interview.
    - Areas for improvement: Suggest areas where the candidate could improve.
    - Technical skills: Evaluate the candidate's
technical skills based on their responses.
    - Communication skills: Assess the candidate's communication skills.
    - Recommendations: Provide specific recommendations for the candidate.
`;
```

While this approach works well in ideal scenarios, improvements could be made to enhance the feedback's depth, conciseness, and focus.

Specifically, the agent could be guided to provide more detailed and targeted feedback while also being trained to ignore irrelevant parts of the conversation to ensure a more streamlined and relevant assessment, implementing a more guided process. There are many opportunities for improvement here, including

- *Incorporating a knowledge base*—Provide the AI with access to a knowledge base of best practices, common interview questions, and expected answers for the specific role. This will allow the AI to provide more accurate and relevant feedback.